A Detection Algorithm of Rail Surface Defects Based on Deep Learning

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***Abstract*—In the field of railway transportation, the integrity of the rails is a core element in ensuring the safe operation of trains. As the railway network continues to expand and the transportation load increases, the detection of surface defects on the rails becomes particularly critical. Neglecting defects such as cracks, wear, and corrosion could lead to catastrophic consequences. This study is dedicated to developing an advanced rail surface defect detection algorithm based on artificial intelligence, significantly enhancing the accuracy and efficiency of detection. We propose an innovative "Enhanced Feature Fusion U-Net" (EFF-U-Net) model, which, through its designed skip connections, achieves efficient fusion of features, thereby greatly improving the recognition ability of the model for details on the rail surface. Faced with the scarcity of rail surface defect datasets, we have employed data augmentation techniques to effectively expand the sample size and enhance the model's generalization capabilities. Through experimental validation on the RSDDS+ dataset, our method has demonstrated exceptional performance, significantly improving the accuracy and identification capability of defect detection.**

Keywords- rail surface defects; deep learning; defect detection; U-Net

I. INTRODUCTION

With the rapid expansion of the railway network and the increasing demand for transportation, rails, as a key infrastructure of the railway system, directly affect the safety and stability of train operations. Defects on the surface of the rails, such as cracks, wear, and corrosion, if not detected and dealt with in a timely manner, can lead to serious accidents, resulting in casualties and property damage, and increasing the maintenance costs of railway operations. The detection of surface defects on rails plays a crucial role in ensuring the safety of railway transportation, improving transportation efficiency, and reducing maintenance costs [1][2]. As artificial intelligence technology continues to spread, Zheng D [3] and Hui Luo [4] have both improved the YOLOv5s to achieve surface defect detection of steel rails. Wang H [5] proposed a surface defect detection network based on Mask R-CNN [6], and experimental results have shown that the designed method can effectively achieve quantitative detection of defects and can effectively segment the boundary information of defects. In order to achieve more accurate surface defect detection of rails, researchers have utilized U-Net [7] series models to detect defects on the rail surface. Wu Y and others proposed a hybrid loss composed of binary cross-entropy, structural similarity (SSIM), and IOU, which has shown good performance in the designed network.

Due to the limited amount of the original rail surface defect dataset, data augmentation methods have been used to expand the sample size based on public datasets. Furthermore, to address the limitations of the original U-Net model in defect detection, an improved model has been designed to adapt to the detection of surface defects on rails.

II. THE METHOD

A. Algorithm design framework

Fig.1 demonstrates the algorithm framework designed in this paper, where (c): three data augmentation techniques are applied to the input images: Random Crop, Random Flip, and Random Shift. These techniques help to improve the model's generalization capabilities. (d) The encoder gradually extracts image features through multiple convolutional layers (Conv), batch normalization (BN), and the ReLU activation function, and reduces the size of the feature maps through max pooling. The decoder then gradually restores the image size through upsampling and convolutional operations, while integrating the feature maps from the encoder (via skip connections) to enhance the precision of segmentation. By combining data augmentation techniques with a complex network structure, the aim is to improve the performance of the defect image segmentation task.

$$crop(I) = I[y : y+h, x : x+w] \tag{1}$$

$$\operatorname{flip}(I)=\begin{cases} I[:,::-1], \text{if horizontal flip} \\ I[::-1,:], \text{if vertical flip} \end{cases} \quad (2)$$

$$\operatorname{shift}(I)=I[y+\Delta y: y+h+\Delta y, x+\Delta x: x+w+\Delta x] \quad (3)$$

B. *The basic principle of U-Net*

The U-Net is an exceptional model for image segmentation tasks, designed to perform pixel-level classification effectively. Its operational process is divided into three main stages: (a) Feature Extraction: During this stage, image features are extracted using a series of convolutional layers followed by max pooling layers. (b) Feature Fusion: After feature extraction, the U-Net moves into the feature fusion phase. The network employs upsampling to enlarge the feature maps, simultaneously integrating features from the corresponding layers of the encoder. (c) Prediction: Finally, the U-Net utilizes the integrated feature layers to classify each pixel within the image.

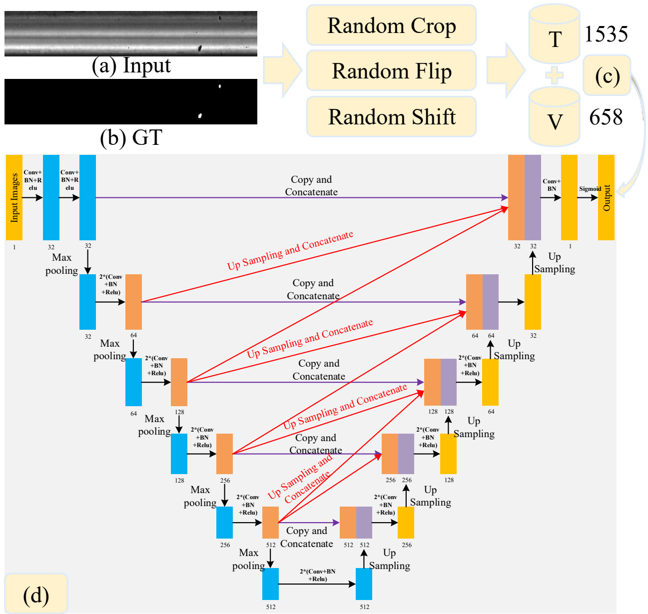

Figure 1. Algorithm frame diagram.

C. *EFF-U-Net*

Fig.2 shows the improved U-Net convolutional neural network architecture designed in this paper for defect segmentation, which we can name the "Enhanced Feature Fusion U-Net" (EFF-U-Net). This model comprises two main parts: the encoder and the decoder, along with skip connections between them. The encoder extracts features through convolutional layers, batch normalization, and ReLU activation functions, followed by downsampling using max pooling layers. The decoder then progressively restores the size of the feature maps through upsampling layers, integrating with the feature maps from the encoder via skip connections to preserve more spatial information and enhance segmentation accuracy. The red lines in the figure represent the innovative aspect of this paper, which is the addition of extra skip connections between the encoder and decoder. These connections effectively merge features from different levels through copying, concatenating, and performing upsampling and convolution operations, thereby enhancing the model's ability to capture details. Ultimately, the output of the decoder generates an accurate segmentation map through a convolutional layer and a Sigmoid activation function, making it suitable for defect segmentation tasks involving complex backgrounds and small objects.

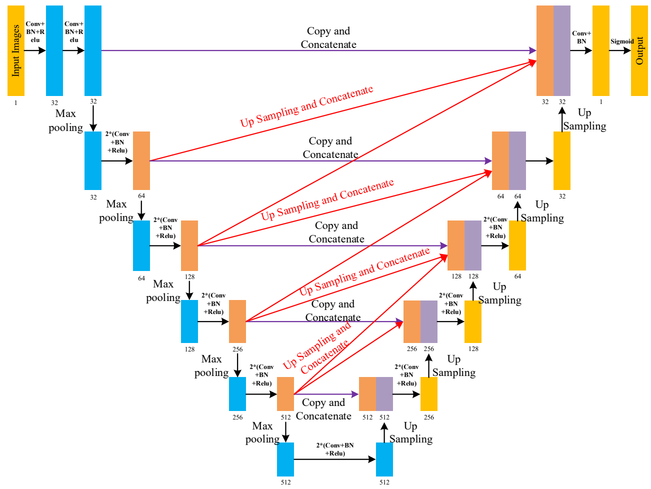

Figure 2. EFF-U-Net algorithm frame.

D. *Loss Fuction*

The Dice Loss is capable of effectively mitigating the adverse effects caused by the imbalance in the area of foreground and background in the samples. This loss function focuses more on the identification of the foreground area during the training process, aiming to reduce the number of False Negatives (FN), which means decreasing the instances of missed detections [8]. However, the Dice Loss may encounter the issue of loss saturation, where the loss value may not decrease significantly during the later stages of training, thereby affecting the model's further optimization. Therefore, relying solely on the Dice Loss may not achieve the desired training outcomes, and it is often necessary to combine it with other loss functions to enhance the model's overall performance.

$$Dice=\frac{2|G \cap P|}{|G|+|P|} \quad (4)$$

$$\ell_{Dice}=1-Dice=1-\frac{2|G \cap P|}{|G|+|P|} \quad (5)$$

III. Experimental Results And Analysis

A. *Experimental configuration*

To construct and train the deep learning model, we have selected a high-performance computing platform featuring an RTX 3050 Laptop graphics card. This platform is powered by an AMD Ryzen 5 5600H processor with Radeon Graphics, operates at a speed of 3.30 GHz, and is endowed with 32.0 GB of memory, offering ample computational resources for the training process. We have opted for the PyTorch deep learning framework for its robustness and flexibility.

For the training parameters, we have set the input size to 320×320 pixels, which is a common resolution that balances the detail of the images with the computational load. The batch size, which determines the number of samples processed before the model is updated, is set to 1. This choice is often made to provide a more precise gradient estimation and to handle the limited memory capacity effectively. The learning rate, a hyperparameter that controls how much the model updates its weights with respect to the loss gradient, is set to 0.001, which is a standard starting point for many deep learning models. Lastly, the number of training steps, which refers to the number of times the model will iterate over the entire dataset, is set to 150. This number can be adjusted based on the model's performance and the available computational resources.

B. Data set

The RSDDS dataset utilized in this paper encompasses images from two distinct sources:

(1) Type I RSDDs dataset: This consists of 67 images captured from high-speed tracks.

(2) Type II RSDDs dataset: It comprises 128 images captured from standard or heavy-duty transportation tracks.

Given the scarcity of additional public datasets, direct training of a deep learning model is challenging. Consequently, this paper employs data augmentation techniques to expand the base datasets, thereby enhancing the model's training process and improving its ability to generalize from the available data, as shown in Fig.3.

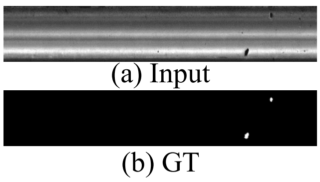

Figure 3. Defect sample.

C. Evaluating Indicator

Precision: The higher the accuracy rate, the more accurate the search.

Recall: The higher the recall rate, the more complete the search.

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

TP: true positive, FN: false negative, FP: false positive, G: true value, P: predicted value.

D. Quantitative Analysis

Table 1 presents a comparison of the Precision and Recall metrics for various algorithms across two datasets: RSDDS and RSDDS+. The data reveals that all algorithms demonstrate superior performance on the RSDDS+ dataset. This improvement is attributed to the enhancements and optimizations applied to the RSDDS+ dataset, which enable the models to learn and identify defects more effectively.

Specifically, the UNet algorithm achieves a precision of 0.7609 and a recall rate of 0.6552 on the RSDDS dataset. These metrics show improvement on the RSDDS+ dataset, where the precision and recall are elevated to 0.7844 and 0.7629, respectively. The U2NetP [9] algorithm exhibits relatively weaker performance on the RSDDS dataset, with a precision of 0.7254 and a recall of 0.8097. However, its performance on the RSDDS+ dataset is notably better, with precision and recall increasing to 0.7534 and 0.7942, respectively.

It is particularly noteworthy that the algorithm designed in this paper outperforms the others on the RSDDS+ dataset, achieving precision and recall values of 0.7846 and 0.8228, respectively. These results indicate that the algorithm proposed in this study possesses greater accuracy and capability in defect identification when applied to the RSDDS+ dataset.

TABLE I. COMPARISON OF ALGORITHM ACCURACY IN THIS PAPER

Method	Dataset	Precision	Recall
UNet	RSDDS	0.7609	0.6552
UNet	RSDDS+	0.7844	0.7629
U2NetP	RSDDS	0.7254	0.8097
U2NetP	RSDDS+	0.7534	0.7942
OUR	RSDDS+	**0.7846**	**0.8228**

E. Comparative Qualitative Analysis

Fig.4 presents a qualitative comparison of the results for rail surface defect detection. Each group in the figure consists of three parts: (a) Input, which shows the original rail images; (b) GT, standing for Ground Truth, the annotated images of rail defects where the white areas indicate the locations of defects; and (c) OUR, which demonstrates the detection results of the Enhanced Feature Fusion U-Net (EFF-U-Net) model proposed in this study.

It can be observed from the figure that our model is capable of accurately identifying defects on the rail surface, such as cracks and wear. In the first and second groups of images, the detection results of the model closely match the Ground Truth, with the defective areas clearly marked. In the third group of images, even though the defects are small and not easily noticeable, the model is still able to accurately pinpoint the defect areas. This indicates the model's effectiveness and robustness in dealing with complex backgrounds and subtle defects.

The results substantiate the efficacy of the EFF-U-Net model in the task of rail surface defect detection and its potential to provide strong technical support for railway transportation safety. By comparing with the Ground Truth, it is evident that the model maintains a high level of accuracy in detecting defects of various types and sizes, which is significant for achieving automated and efficient rail maintenance.

Fig.5 demonstrates the effect of defect detection on railway images with a large aspect ratio. Each group in the figure consists of three parts: (a) Input, which represents the original railway images input into the detection system; these images have a large aspect ratio, increasing the complexity for the detection algorithm to process; (b) GT, standing for Ground Truth, where the white areas indicate the locations of actual defects; and (c) OUR, which shows the detection results of the algorithm proposed in this paper.

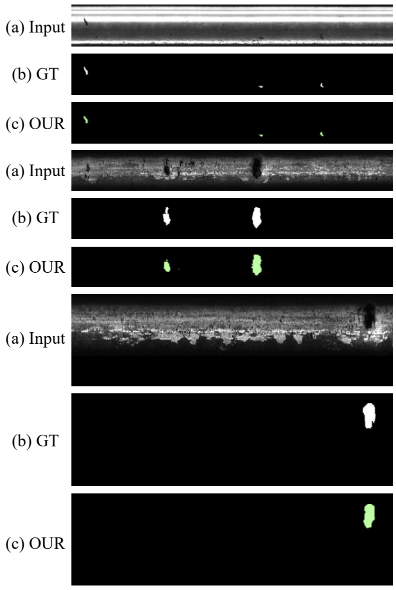

Figure 4. Defect detection effect.

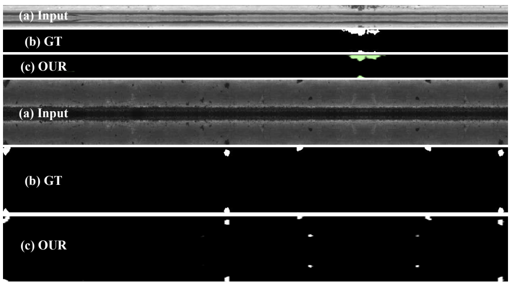

Figure 5. The effect of defect detection on railway images with a large aspect ratio.

From Fig.5, it can be observed that despite the large aspect ratio of the input images, which could potentially make defects appear smaller and more dispersed, thus increasing the difficulty of detection, the algorithm is still capable of effectively identifying defects on the railway surface. In the first set of images, the algorithm successfully detected the defect area located in the middle of the image. In the second set, the algorithm also accurately marked several scattered defect points. The results indicate that the algorithm in this study has good defect detection capabilities even when dealing with images of large aspect ratios, providing reliable technical support for railway transportation safety.

Fig.6 shows a qualitative comparison of the detection results for rail surface defects, highlighting an instance where the detection was inaccurate. The caption "Defect detection effect (error)" suggests a discrepancy between the actual annotations and the algorithm's output. Observing Fig.6, it is evident that the algorithm did not accurately detect the defects as annotated in the Ground Truth (GT). The white areas, which signify defects in the GT, are not adequately represented in the OUR section, implying that the model may have overlooked these defects or is less sensitive to certain types of defects. To enhance the model's accuracy in detecting such defects, further analysis and refinement of the model or its training process may be warranted.

Figure 6. The effect of defect detection on railway images with a large aspect ratio(error).

IV. Conclusions

In this paper, we propose a novel rail surface defect detection algorithm by enhancing an existing deep learning model. The addition of skip connections in the EFF-U-Net model enables it to effectively integrate features from various levels, thereby enhancing the precision of defect segmentation. The experimental results, particularly on the RSDDS+ dataset, demonstrate the superior performance of this method in addressing rail surface defect detection tasks, confirming its effectiveness. Furthermore, the utilization of data augmentation techniques contributes significantly to bolstering the model's generalization capabilities, which is especially beneficial when dealing with limited datasets. The research presented in this paper offers an innovative solution for detecting defects on rail surfaces, which can contribute to enhancing the safety and efficiency of railway transportation systems.

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